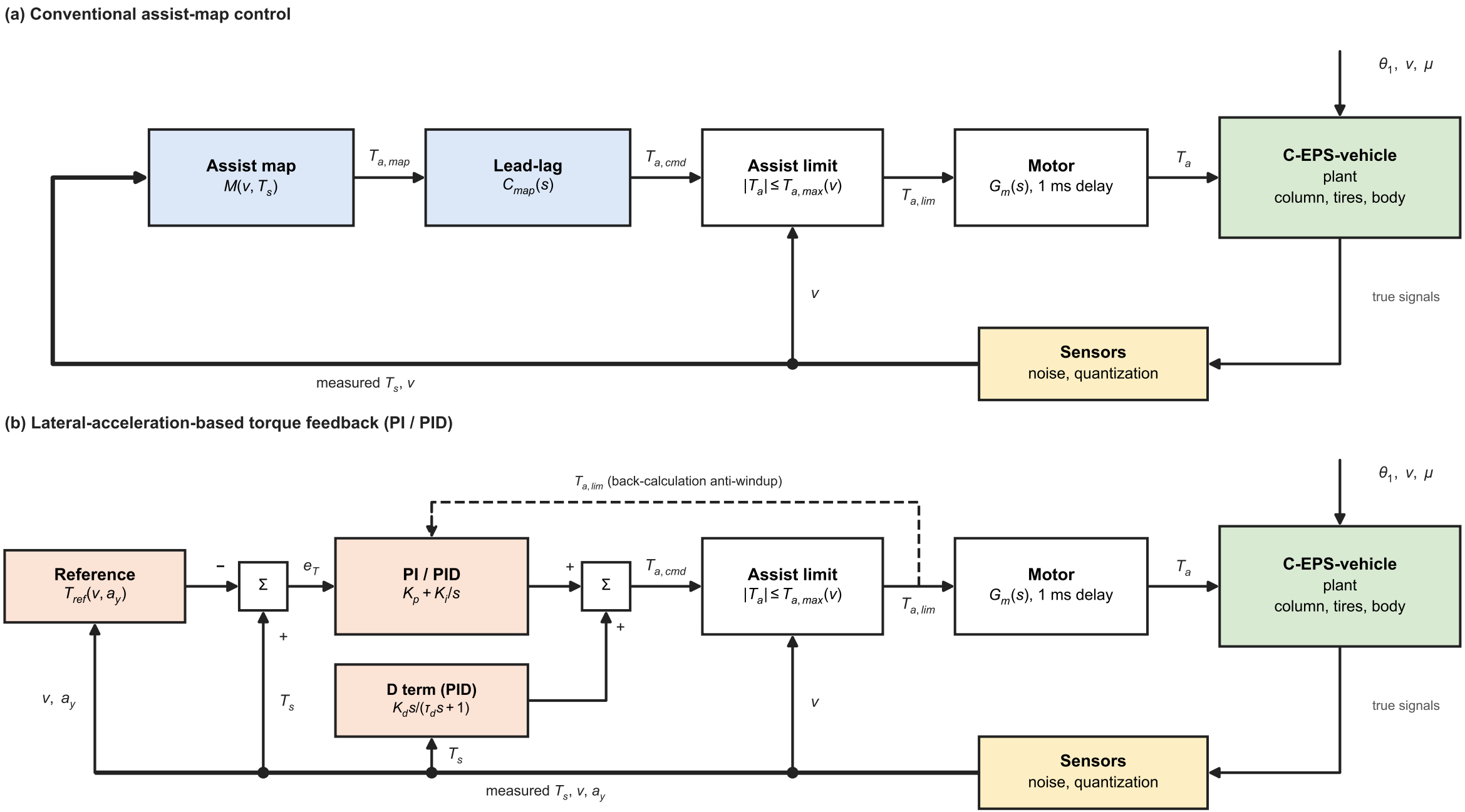
(a) Conventional assist-map control
θ1, v, μ
Assist map
M(v, Ts)
Ta, map
Lead-lag
Cmap(s)
Ta, cmd
Assist limit
|Ta| ≤ Ta, max(v)
Ta, lim
Motor
Gm(s), 1 ms delay
Ta
C-EPS-vehicle
plant
column, tires, body
true signals
v
Sensors
noise, quantization
measured Ts, v
(b) Lateral-acceleration-based torque feedback (PI / PID)
Ta, lim (back-calculation anti-windup)
θ1, v, μ
Reference
Tref(v, ay)
Σ
eT
PI / PID
Kp + Ki/s
Σ
Ta, cmd
Assist limit
|Ta| ≤ Ta, max(v)
Ta, lim
Motor
Gm(s), 1 ms delay
Ta
C-EPS-vehicle
plant
column, tires, body
true signals
v, ay
Ts
D term (PID)
Kds/(τds + 1)
v
Ts
Sensors
noise, quantization
measured Ts, v, ay